

Special Collection: Twenty Years of West Nile Virus in the United States

Impact of West Nile Virus on Bird Populations: Limited Lasting Effects, Evidence for Recovery, and Gaps in Our Understanding of Impacts on Ecosystems

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Abstract

The introduction of West Nile virus to North America in 1999 had profound impacts on human and wildlife health. Here, we review studies of WNV impacts on bird populations and find that overall impacts have been less than initially anticipated, with few species showing sustained changes in population size or demographic rates across multiple regions. This raises four questions: 1) What is the evidence for WNV impact on bird populations and how can we strengthen future analyses? We argue that future studies of WNV impacts should explicitly incorporate temporal variation in WNV transmission intensity, integrate field data with laboratory experimental infection studies, and correct for multiple comparisons. 2) What mechanisms might explain the relatively modest impact of WNV on most bird populations? We suggest that spatial and temporal variation in WNV transmission moderates WNV impacts on species that occur in multiple habitats, some of which provide refugia from infection. 3) Have species recovered from the initial invasion of WNV? We find evidence that many species and populations have recovered from initial WNV impact, but a few have not. 4) Did WNV cause cascading effects on other species and ecosystems? Unfortunately, few studies have examined the cascading effects of WNV population declines, but evidence suggests that some species may have been released from predation or competition. We close by discussing potentially overlooked groups of birds that may have been affected by WNV, and one highlight species, the yellow-billed magpie (*Pica nuttalli* Audubon, 1837 [Passeriformes: Corvidae]), that appears to have suffered the largest range-wide impact from WNV.

Key words: vector-borne pathogen, avian infection, biodiversity, ecology and population biology, West Nile virus

The introduction of West Nile virus (WNV) into North America in 1999 resulted in both an outbreak of encephalitis in humans and horses and mortality in both wild and captive birds (Lanciotti et al. 1999). The subsequent spread across North America was tracked, in part, by testing dead wild birds submitted to health agencies for WNV infection, and these dead birds provided a useful predictor of human cases (Gupitill et al. 2003, Reisen et al. 2006, Nielsen and Reisen 2007). The enormous numbers of dead corvids that tested positive for WNV, as well as many other species, quickly raised the question of how WNV might affect bird populations as it spread across the Americas (Bernard and Kramer 2001, Caffrey and Peterson 2003, Yaremych et al. 2004, Reisen et al. 2006, Kilpatrick et al. 2007, Wheeler et al. 2009, Kwan et al. 2012a).

There have been four large-scale (statewide or larger) studies of WNV impacts on population trends of a broad suite of bird species. These studies compared population counts before the arrival of

WNV in 1999 with counts afterward. These studies took advantage of the extremely valuable rich, and in the case of the breeding bird survey (BBS) data, standardized censusing methodology (BBS data include >4100 routes in North America, with each route composed of 50 census points separated by 0.5 miles; Sauer et al. 2017, Pardieck et al. 2018). These studies included analyses of Christmas bird counts through 2002 in the northeastern United States (Caffrey and Peterson 2003); BBS counts across the state of California through 2007 (Wheeler et al. 2009), BBS counts across 10 states spanning the United States through 2005 (LaDeau et al. 2007), and BBS counts in 10 states through 2010 (Foppa et al. 2011). There have been several additional studies examining population trends of individual species, including ruffed grouse (*Bonasa umbellus* Linnaeus, 1766 [Galliformes: Phasianidae]) (Stauffer et al. 2018), yellow-billed magpie (*Pica nuttalli*) (Crosbie et al. 2008), and American crow (*Corvus brachyrhynchos* Brehm, 1822) (LaDeau et al. 2011). There also have been studies of

WNV impacts on survival of individual species using either mark-recapture or radio-telemetry, including house finch (*Haemorrhous mexicanus* Müller, 1776 [Passeriformes: Fringillidae]) (Pellegrini et al. 2011), sage grouse (*Centrocercus urophasianus* Bonaparte, 1827 [Galliformes: Phasianidae]) (Naugle et al. 2004), eastern bluebird (*Sialia sialis* Linnaeus, 1758 [Passeriformes: Turdidae]) (Hill et al. 2010), and American crow (Yaremych et al. 2004, Ward et al. 2006), and one study examined differences in survival pre- and post-WNV arrival in 49 species of birds (George et al. 2015). Despite the enormous numbers of dead birds testing positive for WNV, strong consistent impacts of WNV on population trends in these studies were limited to few species, and only for American crows and yellow-billed magpies were population impacts evident in all or nearly all regions examined (LaDeau et al. 2007, Wheeler et al. 2009). Similarly, strong, persistent, and sizeable effects on annual survival that could be attributed to WNV were limited to just a few species, despite claims of more widespread effects (George et al. 2015). Large widespread impacts on American crows are likely due to the high mortality following infection in this species (~100% in laboratory experimental infection studies; Komar et al. 2003; Brault et al. 2004, 2007), overutilization by mosquitoes relative to crow abundance (Kilpatrick et al. 2006a), and possibly the wide-ranging movements of individuals that bring them in contact with WNV-infectious mosquitoes (Ward et al. 2006).

Two potential mechanisms that could explain the limited WNV impacts are spatial and temporal variation in WNV transmission. If a species occurs across a range of habitats, and WNV transmission is lower in one of these habitats, then this area may serve as a refuge from infection. For example, several studies have shown that WNV transmission is higher in urban or agricultural areas than undeveloped habitats (Andreadis et al. 2004, Bradley et al. 2008, Gómez et al. 2008, Bowden et al. 2011, Kilpatrick 2011, Kovach and Kilpatrick 2018). If species occur both in urban and agricultural areas and in undisturbed habitats (e.g., forest), populations in less-disturbed habitats may be source populations that produce immigrants that repopulate urban and agricultural habitats where WNV transmission appears to be higher (LaDeau et al. 2011). Most previous studies of WNV impacts have examined trends at large scales (e.g., the statewide level), whereas finer-scale analyses (e.g., using individual BBS routes, rather than statewide aggregations) might uncover much larger impacts in smaller areas where WNV transmission is especially intense. Temporal variation in the intensity of WNV transmission, which is extensive (Reisen et al. 2008, Kilpatrick and Pape 2013), would also modulate impacts and allow recovery in low transmission years.

The limited, spatiotemporally variable differences in bird population counts before and after WNV arrival raises four questions. 1) How strong is the existing evidence for WNV impact on bird populations and how can we improve future analyses? 2) What mechanisms might explain the relatively modest impact of WNV on bird populations? 3) Have species recovered from the initial invasion of WNV? 4) Did WNV cause cascading effects on other species and ecosystems?

Evidence for WNV Impacts

Approaches taken in past studies have probably led to both over- and underestimates of WNV impacts. Most studies have made comparisons of avian population abundance, trends, or demographic rates (e.g., survival) between years before and after WNV arrival (see references above). Many studies have examined multiple species and made large numbers of comparisons without correcting for the inflated probability of detecting WNV impacts when they are absent (e.g., George et al. 2015) examined 49 species and compared

six possible WNV effects for a total of nearly 300 comparisons; LaDeau et al. (2007) examined 20 species in 6 regions and 1–6 yr post-WNV-arrival independently for >400 comparisons). None of the published studies have incorporated direct metrics of temporal variation in WNV transmission intensity (e.g., the density of infected mosquitoes), which have been shown to be strongly correlated with the number of human WNV cases (Bolling et al. 2009, Kwan et al. 2012b, Kilpatrick and Pape 2013, Paull et al. 2017). This is important because WNV transmission varies enormously from year to year, due to weather, avian immunity, vector competence, viral evolution, and other factors (Kilpatrick et al. 2008, 2010; Reisen et al. 2008; Kwan et al. 2012a; Chung et al. 2013; Paull et al. 2017), and therefore one would expect higher WNV impacts in some years than in others. Some studies have used the number of human WNV cases per year as a surrogate for transmission intensity either directly or indirectly (Caffrey and Peterson 2003; LaDeau et al. 2007, 2011; Foppa et al. 2011), which is a substantial improvement over simple pre-post comparisons, but is suboptimal because human immunity following large WNV epidemics greatly reduces the number of subsequent cases and partly obscures the relationship between human cases and the density of infected mosquitoes (Paull et al. 2017).

One way to strengthen evidence for WNV impacts would be to combine observational studies of population counts with evidence for mortality from experimental infection studies. For example, WNV experimental infection studies demonstrating substantial mortality (15–100% of infected birds) in American crow, blue jay (*Cyanocitta cristata* Linnaeus, 1758 [Passeriformes: Corvidae]) black-billed magpie (*Pica hudsonia* Sabine, 1823 [Passeriformes: Corvidae]) (Komar et al. 2003), California scrub-jay (*Aphelocoma californica* Vigors, 1839 [Passeriformes: Corvidae]) and house finch (Reisen et al. 2005), American robin (*Turdus migratorius* Linnaeus, 1766 [Passeriformes: Turdidae]) (VanDalen et al. 2013), ruffed grouse (Nemeth et al. 2017), and tufted titmouse (*Baeolophus bicolor* Linnaeus, 1766 [Passeriformes: Paridae]) (Kilpatrick et al. 2013) have provided supporting evidence to corroborate population declines coincident with WNV arrival in these species. One can build even stronger evidence for WNV being the cause of population or demographic deviations by generating predictions of WNV impact using experimental infection studies and patterns of exposure to WNV either based on mosquito feeding studies or serology (LaDeau et al. 2007, Wheeler et al. 2009). Several studies have quantified feeding preferences or host utilization indices (Hassan et al. 2003, Kilpatrick et al. 2006a, Hamer et al. 2009, Kent et al. 2009, Janousek et al. 2014), and these can be used to predict relative exposure risk.

A second way to improve future analyses would be to use an explicit, temporally variable index of WNV transmission as the predictor in analyses, rather than individual comparisons of pre-post population counts or survival estimates from individual years. This would greatly reduce the number of comparisons made and reduce the likelihood of type II statistical errors (falsely attributing a low count or survival estimate to WNV). As noted above, the ideal index of WNV transmission would be the density of infected mosquitoes, and data to estimate this is collected by nearly all vector control agencies. Failing this, data on yearly WNV incidence at the county scale are publicly available (Centers for Disease Control and Prevention 2019). These temporally variable data could be combined with relationships between land use and WNV transmission (Andreadis et al. 2004, Bradley et al. 2008, Gómez et al. 2008, Bowden et al. 2011, Kilpatrick 2011, Kovach and Kilpatrick 2018) to generate a spatiotemporal predictor of WNV at relatively fine spatial scales (e.g., a single BBS route) to provide the most direct analysis of WNV impacts. Regardless of which measure of WNV transmission is used,

studies should always correct for multiple comparisons made, especially when multiple species, regions, or years are examined.

Spatiotemporal Variability in WNV Impacts and Species' Recoveries

In all past studies, WNV impacts on populations have been variable across both space and time. For many species, anomalously low populations or survival estimates were limited to single years or regions, which raises the question of whether populations have recovered from initial WNV impacts. Although a formal analysis of population dynamics is beyond the scope of this forum paper, data from the Breeding Bird Survey (<https://www.pwrc.usgs.gov/BBS/RawData/>; Sauer et al. 2017, Pardieck et al. 2018), corrected for survey effort (the number of BBS routes counted each year), offer some insights into broad-scale patterns of population trends and indicate directions for future research. BBS counts are done each year in June, before peak WNV transmission, so population impacts are likely to be most evident starting with the following year's count. Our goal was to crudely assess whether species that previous studies had suggested were impacted by WNV had recovered to pre-WNV abundances at the statewide scale and thus we restricted our examination of population recovery to those species.

We first examined American crow populations, which were the most widely impacted species in past studies (LaDeau et al. 2007, 2011; Foppa et al. 2011). Statewide counts of American crow populations initially declined with WNV arrival in New York, Maryland, Virginia, and Illinois between the year of WNV arrival and ~2005, but declines were less evident in Colorado and Oregon (Fig. 1). Partial recovery appears to have occurred in some of these populations (e.g., New York, Maryland, Virginia), but not others (Illinois), and some crow populations declined again between 2011 and 2014 for unknown reasons. However, all four of the states where declines were most evident have crow abundances that are now lower than in the late 1990s before WNV was introduced. The net effect of WNV on crow populations thus appears to have been to stop the growth that occurred during the 1970s–1990s. Populations seem to be fluctuating around a new equilibrium that is similar to crow abundance in each state in the 1970s or 1980s before populations grew substantially.

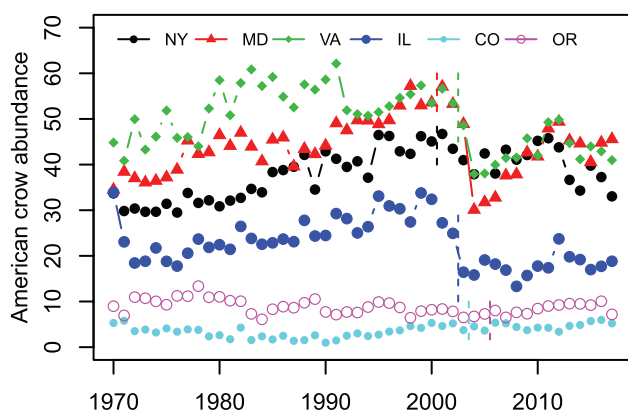


Fig. 1. American crow abundance (in birds per BBS route) in six states spanning North America. WNV arrived statewide in New York (NY) and Maryland (MD) in 2000, in Virginia (VA) and Illinois (IL) in 2002, in Colorado (CO) in 2003, and in Oregon (OR) in 2005, with WNV impacts expected to be most evident to the right of the vertical dashed lines for each state.

The second species we examined was the California endemic yellow-billed magpie, which previous studies have suggested was at high risk from WNV (Crosbie et al. 2008). This species appears to be on a continuous decline since the arrival of WNV (Fig. 2) and shows no sign of recovery. This species may have experienced greater disease impacts than other species because it has a limited range and WNV transmission has occurred throughout its range in every year since 2004 (Reisen et al. 2009; maps.calsurv.org 2019).

Five other bird species with apparent WNV impacts from one or more studies were American robin, tufted titmouse, house and purple finch (*Haemorhous purpureus* Gmelin, 1789 [Passeriformes: Fringillidae]), and black-headed grosbeak (*Pheucticus melanocephalus* Swainson, 1827 [Passeriformes: Cardinalidae]) (LaDeau et al. 2007, Wheeler et al. 2009, George et al. 2015), with two of these species also suffering very high mortality in experimental infections (tufted titmouse, 100%, Kilpatrick et al. 2013; house finch 67%; Komar et al. 2003, Reisen et al. 2005, Fang and Reisen 2006). Plots of the abundance of these species in multiple states overtime show large fluctuations with some temporary declines that might be attributable to WNV (Fig. 3: e.g., tufted titmouse in Maryland and Virginia [but not New York]; purple finch in California). However, overall the populations were variable and there were few strongly evident patterns, especially compared with population impacts of other introduced pathogens such as white-nose syndrome in bats (Langwig et al. 2012), and chytridiomycosis in amphibians (Vredenburg et al. 2010), which have caused complete population extirpations and region-wide species declines of >90%. Most importantly, for our goal here, all but purple finch in California appear to have recovered from WNV impacts at the state scale (or began declines for unknown reasons after 2010). In summary, WNV impacts on bird populations across North America have been sustained across multiple regions in only one species, American crow, and impacts on other species have been spatiotemporally limited, and most have recovered to previous abundances.

A statistical analysis that incorporated the long-term trends in the populations and that addressed the issues described above (i.e., one that examined populations at finer spatial scales and that used a quantitative metric of WNV transmission intensity as the main predictor of WNV impacts) could more clearly determine the extent of population recovery and identify the fraction of the variation in population fluctuations that is likely due to WNV. In addition, analyses of WNV impacts on bird species that have yet to be examined (e.g., shorebirds, raptors, etc.), or other taxa that may suffer WNV

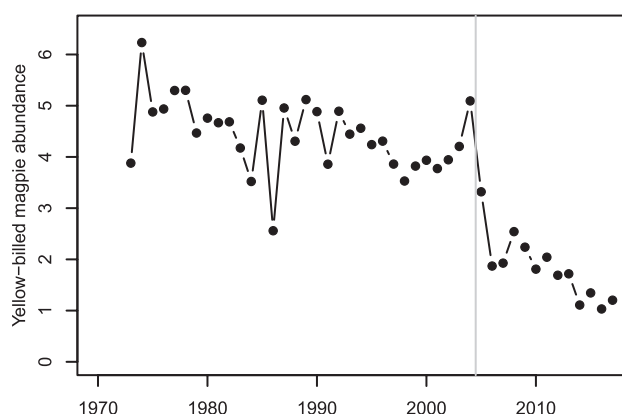


Fig. 2. Yellow-billed magpie abundance in California in birds per BBS route. WNV arrived statewide in California in 2004, with WNV impacts expected to be most evident to the right of the gray vertical line.

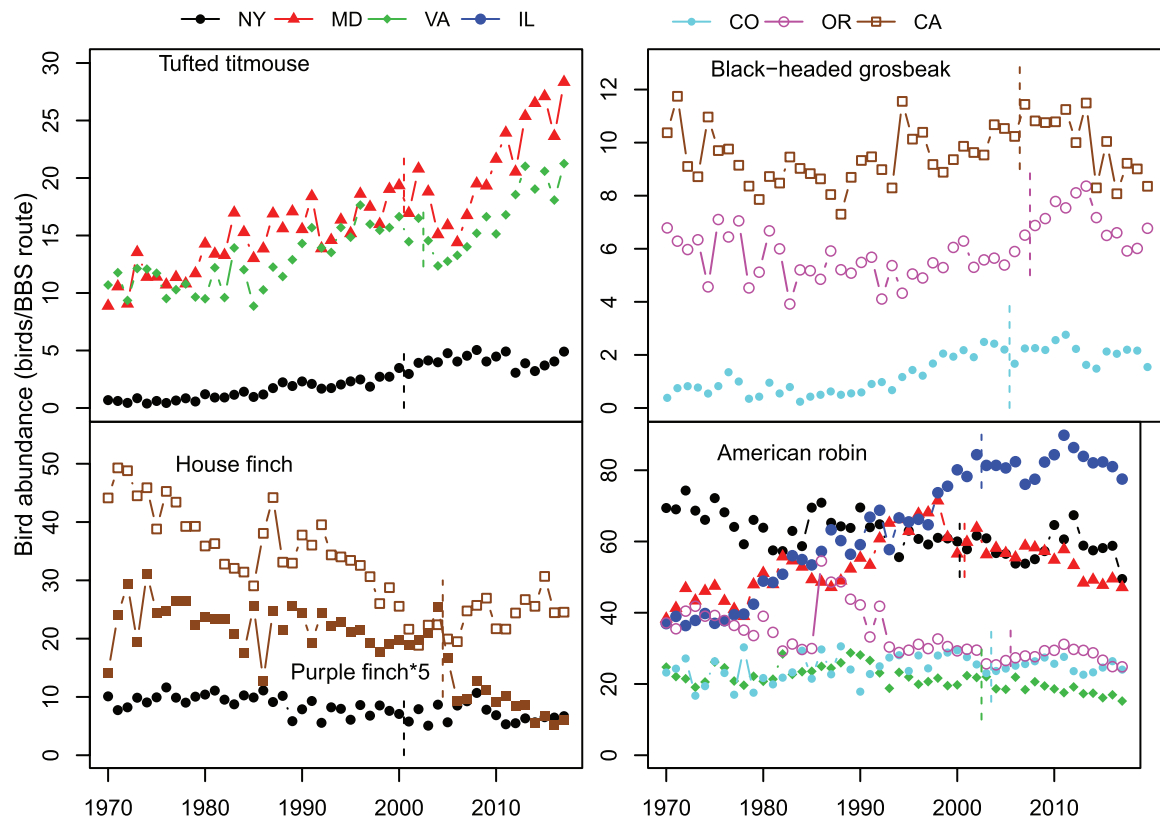


Fig. 3. Tufted titmouse, black-headed grosbeak, house finch (open symbols) and purple finch (closed symbols), and American robin abundance (in birds per BBS route) in three to six states across North America. Purple finch abundance was multiplied by 5 to facilitate display. WNV arrived statewide in New York (NY) and Maryland (MD) in 2000, in Virginia (VA) and Illinois (IL) in 2002, in Colorado (CO) in 2003, in California (CA) in 2004, and in Oregon (OR) in 2005, with WNV impacts expected to be most evident to the right of the vertical dashed lines for each state.

mortality (e.g., American alligators, *Alligator mississippiensis* Daudin, 1802 [Crocodylia: Alligatoridae]) (Klenk et al. 2004) could be fruitful. Finally, no study has examined impacts of WNV on bird or other wildlife populations in Latin America, despite evidence for similar transmission intensities as in parts of North America (Morales-Betoulle et al. 2013), and the presence of many species that are in taxonomic families that suffer high mortality when infected (e.g., corvids; Komar et al. 2003, Reisen et al. 2005).

An outstanding question is what mechanism(s) have allowed species populations to persist with substantial WNV mortality and recover from temporary WNV-related declines. Potential mechanisms include the previous presence or evolution of resistance or tolerance, demographic compensation, changes in virulence due to changes in virus genetics, and reduced WNV transmission intensity. There is little evidence to arbitrate between these mechanisms, but one highly innovative study that measured virus and host traits over time by repeatedly infecting hosts with viruses collected over a decade showed that house sparrows have evolved to be more resistant to the introduced NY99 strain of WNV, whereas the virus has evolved to replicate more efficiently in this host, leading to increased host competence (Duggal et al. 2014). Another study examined temporal variation in WNV seroprevalence in American and fish crows (*Corvus ossifragus* Wilson, 1812 [Passeriformes: Corvidae]) to examine trends in susceptibility to WNV mortality (Reed et al. 2009). However, the data in this study were purely observational and thus are suggestive at best. Additional experimental infection studies of certain species and populations that were both previously used in experimental infection studies, and

are in regions where WNV population impacts were evident (e.g., American crow) could be used to examine changes in resistance or tolerance. In addition, studies that use a range of viral isolates could also examine coevolution between hosts and the virus over time (Duggal et al. 2014).

Cascading Effects of WNV Impacts on Other Species and the Ecosystem

Although sustained regional impacts of WNV on many bird populations were limited to a few species, these declines may have affected other species and the surrounding ecosystem. Unfortunately, there have been few published papers examining the cascading impacts of WNV declines in bird populations. One study examined whether mortality itself might increase WNV transmission (Foppa and Spielman 2007), and changes in the abundance of key species (e.g., American robins) have been linked with mosquito feeding shifts to other species (Edman and Taylor 1968, Kent et al. 2009), sometimes including humans (Kilpatrick et al. 2006b).

One hypothesized impact of a decline in corvid abundance is an increase in nesting success of other bird species because corvids are an important source of nest failure in many species. Another possibility is reduced competition for food or territories between species that are differentially susceptible to WNV mortality (Komar et al. 2003, Reisen et al. 2005) or are differentially fed on by mosquitoes (Kilpatrick et al. 2006a, Hamer et al. 2009, Kent et al. 2009, Thiemann et al. 2011, 2012). There are a few studies quantitatively addressing these possibilities, but anecdotal observations suggest

that these cascading effects may have occurred. Increased capture rates of house sparrows (*Passer domesticus* Linnaeus, 1758 [Passeriformes: Passeridae]) were noted at WNV serosurveillance capture sites during a WNV outbreak in Los Angeles as house finch numbers declined and large numbers of WNV-positive dead American crows were detected (Kwan et al. 2010). House sparrows suffer lower mortality than house finches and are underutilized by mosquitoes relative to their abundance, whereas house finches are preferentially fed upon (Kilpatrick et al. 2006a, Hamer et al. 2009).

One might also hypothesize that a decline in American crows could release other corvids from competition, given that American crows are the most consistently impacted species. Abundance data from two states where American crows (which suffer close to 100% mortality from WNV infection; Komar et al. 2003; Brault et al. 2004, 2007), fish crows (which suffer approximately 50–55% mortality; Komar et al. 2003, Turell et al. 2003), and blue jays (which suffer approximately 75% mortality; Komar et al. 2003) are all at least moderately abundant showed a decline in American crows when WNV arrived statewide (Fig. 4), as previously demonstrated (LaDeau et al. 2007). In Illinois, fish crow populations spiked following the decline of American crows, following a slow gradual rise over the previous two decades. Over the same period, blue jays showed a slight decline with no hint of a competitive release and there was no hint of a release of fish crows in Virginia despite a substantial decline in American crows. A more rigorous analysis would use an actual statistical analysis and examine other factors affecting all three species' population over the last 50 yr, such as land use and climate.

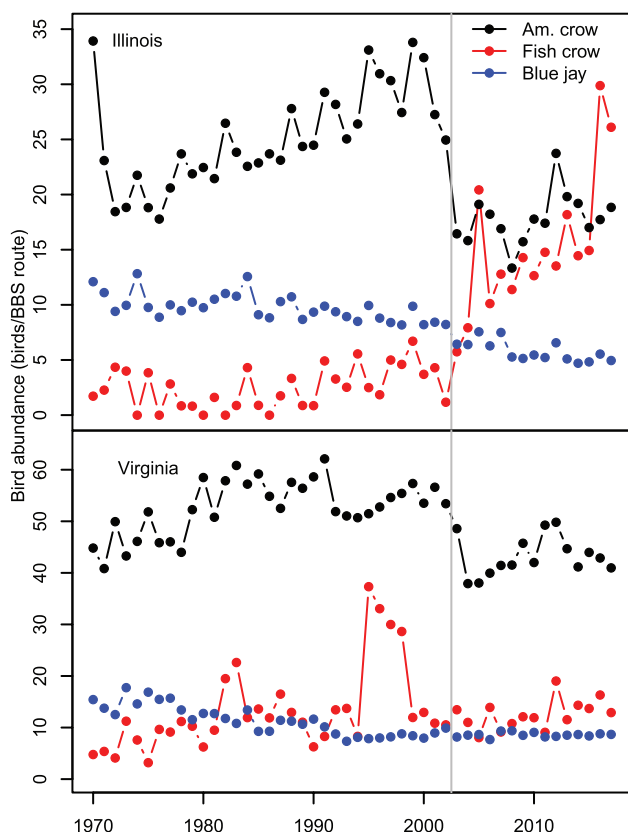


Fig. 4. American crow, fish crow, and blue jay abundance in two states based on regional summaries of Breeding Bird Survey data in birds per BBS route. WNV arrived statewide in Virginia (VA) and Illinois (IL) in 2002, with WNV impacts expected to be most evident to the right of the gray vertical line.

More generally, future work should explore the cascading effects of WNV impacts including predator–prey, competition, and possibly other interactions. In addition, future work should examine the potential for ecosystem consequences of changes in bird abundance, which could include impacts on seed dispersal (e.g., American robins declined in some regions coincident with WNV arrival), insect abundance from declining insectivores (with potential cascading effects on plants), and possible changes in rodent abundance due to decline of raptors (if present), with cascading effects on disease (Levi et al. 2012).

Conclusions

Although WNV has killed large numbers of North American birds, its impact on the avifauna has been smaller than many initially feared, and lasting impacts on population abundance appear to be limited to a few species. A few species of conservation concern have been protected from WNV by vaccination (e.g., island scrub-jays, *Aphelocoma insularis* Henshaw, 1886 [Passeriformes: Corvidae] [Boyce et al. 2011] and California condors, *Gymnogyps californianus* Shaw, 1797 [Accipitriformes: Cathartidae] [Chang et al. 2007]), which is clearly time consuming, logistically challenging, and only suitable for very small and highly threatened populations (Langwig et al. 2015). For other species, such as yellow-billed magpie, one hopes that natural selection will result in birds becoming resistant or tolerant, and efforts to reduce other threats (e.g., predators, land use, etc.) to this and other impacted species can facilitate evolution (Kilpatrick 2006).

Overall, the moderate to low severity of WNV impacts on avian populations is likely due to the enormous spatial variability in WNV transmission intensity, as well as temporal variability that allows recovery of populations. Species that persist across a range of habitats are more likely to have refuges from infection. Over the past two decades, we have learned a great deal about the impacts of an introduced pathogen on wildlife across a continent, and future work has the potential to further refine our understanding.

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